# CSE 242 - Assignment 3

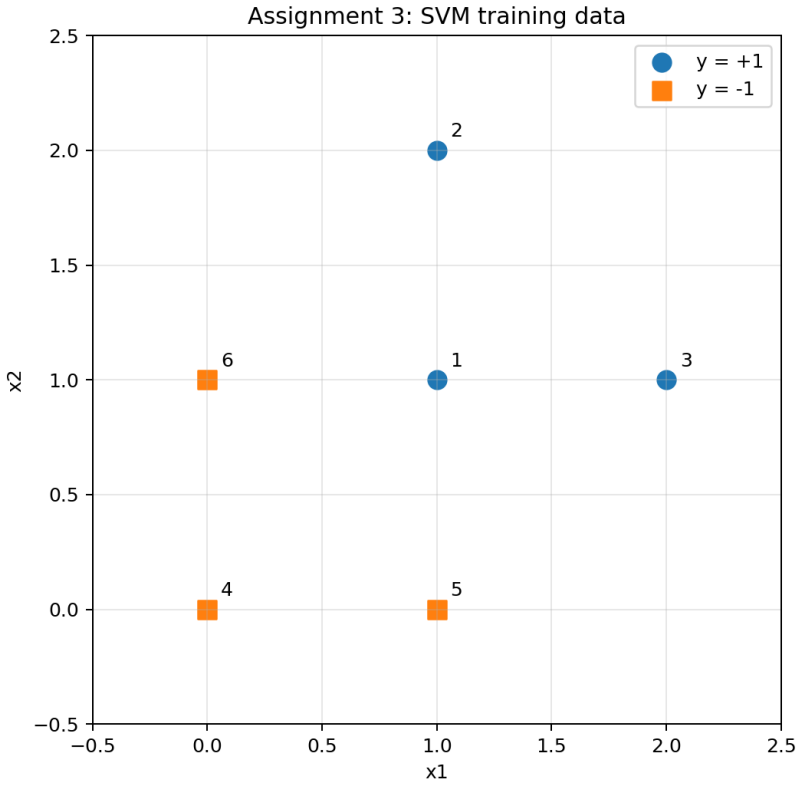
SVMs, K-means, and K-NN | Reproducible experiments: seed 0

# 1. Hard-margin linear SVM

Support vectors are point 1, (1,1), and points 5 and 6, (1,0) and (0,1). They lie on the nearest parallel supporting lines x1+x2=2 and x1+x2=1. All other training points are farther away.

In canonical scaling, support vectors satisfy y(w dot x+b)=1. Thus w1+w2+b=1, w1+b=-1, and w2+b=-1. Subtracting gives w1=w2=2 and b=-3. The separator is 2x1+2x2-3=0, or x1+x2=1.5.

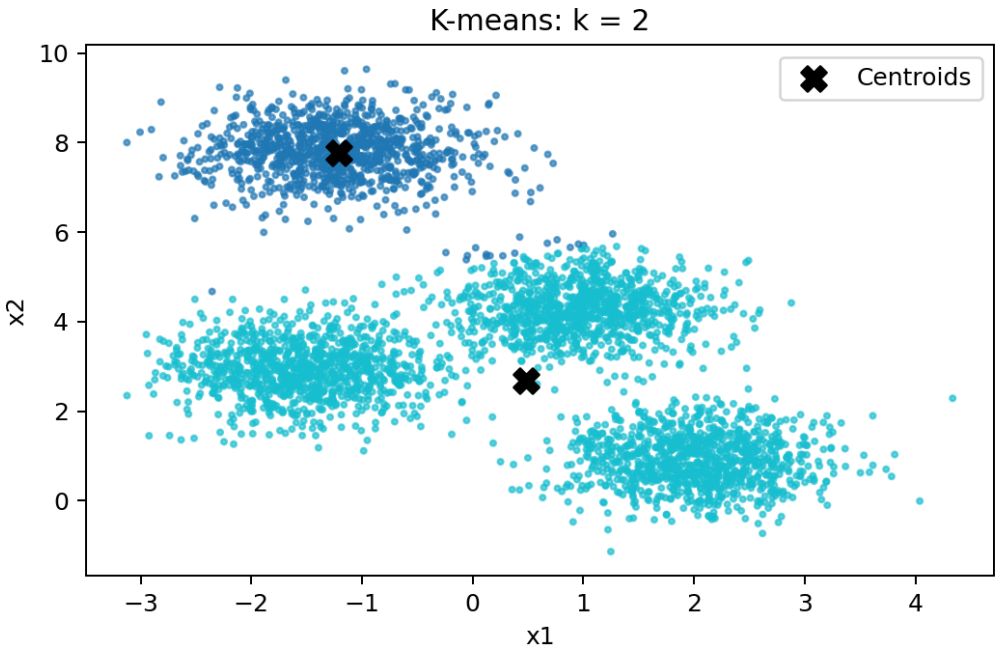
Optimality: the positive convex hull contains (1,1); the negative convex hull contains (0.5,0.5). Their distance is 1/sqrt(2), which bounds the separation between any supporting hyperplanes. The proposed parallel supports attain this bound, proving maximality. The geometric margin is 1/||w||=1/sqrt(8)=0.353553. The full supporting-strip width is 2/||w||=0.707107.

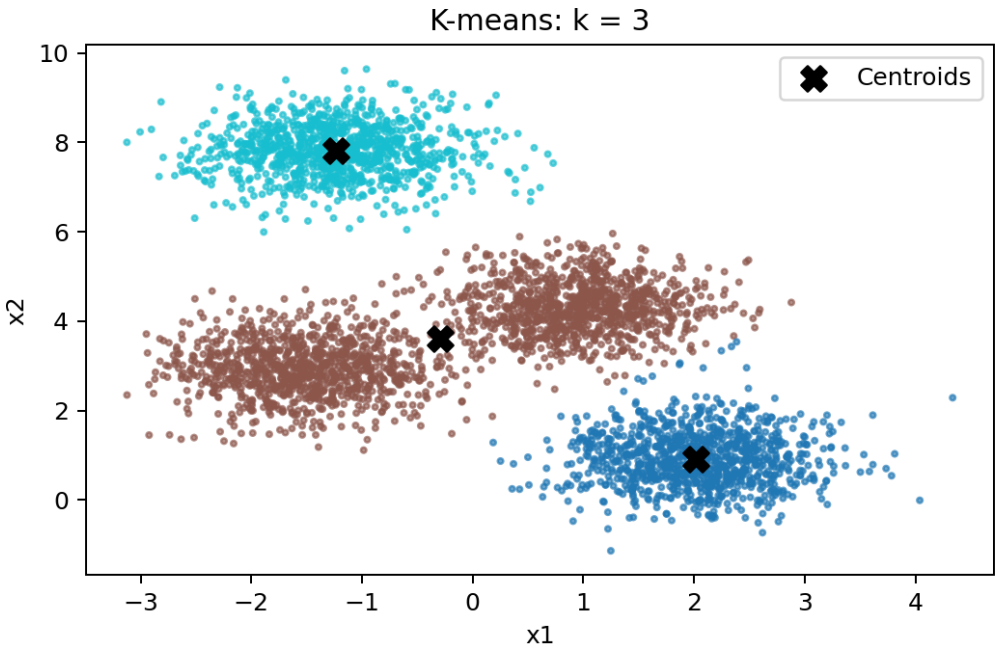


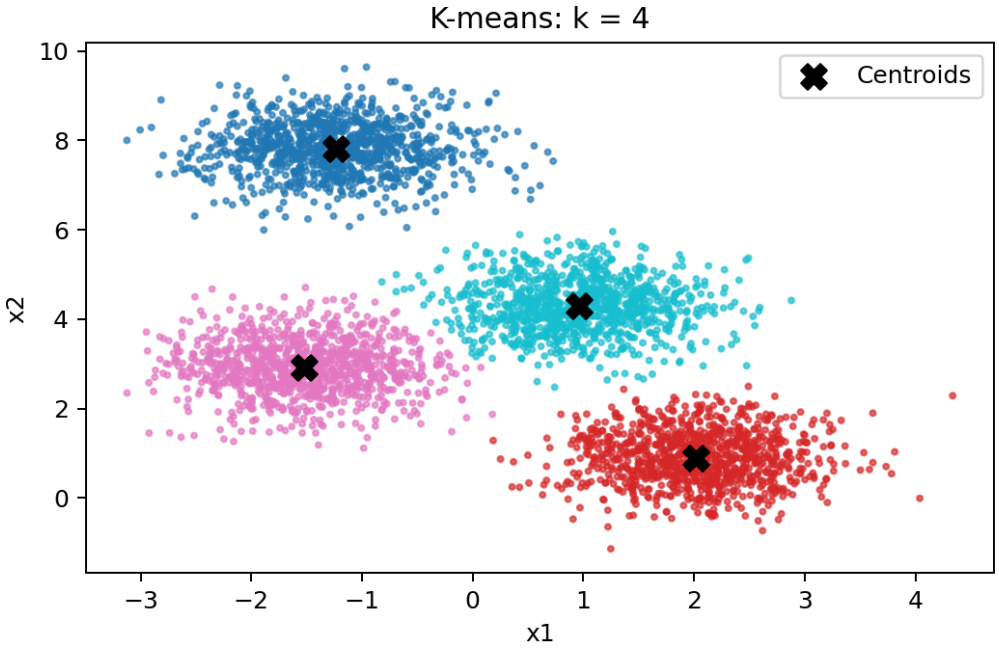
# 2. K-means

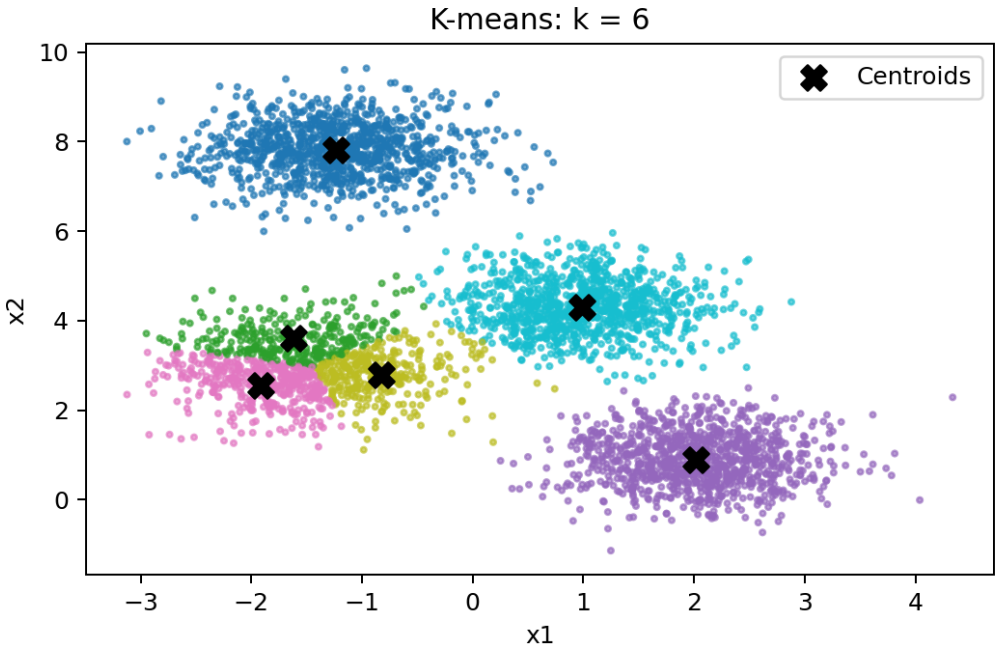
Assignments minimize squared Euclidean distance, with equal distances assigned to the lowest centroid index. Each nonempty cluster is updated to its arithmetic mean. Empty clusters retain their previous centroid; updates copy the input. Initialization samples rows with replacement using one seed-0 generator across k=2,3,4,6,10. Each run performs 100 iterations and a final assignment. True labels are never used.

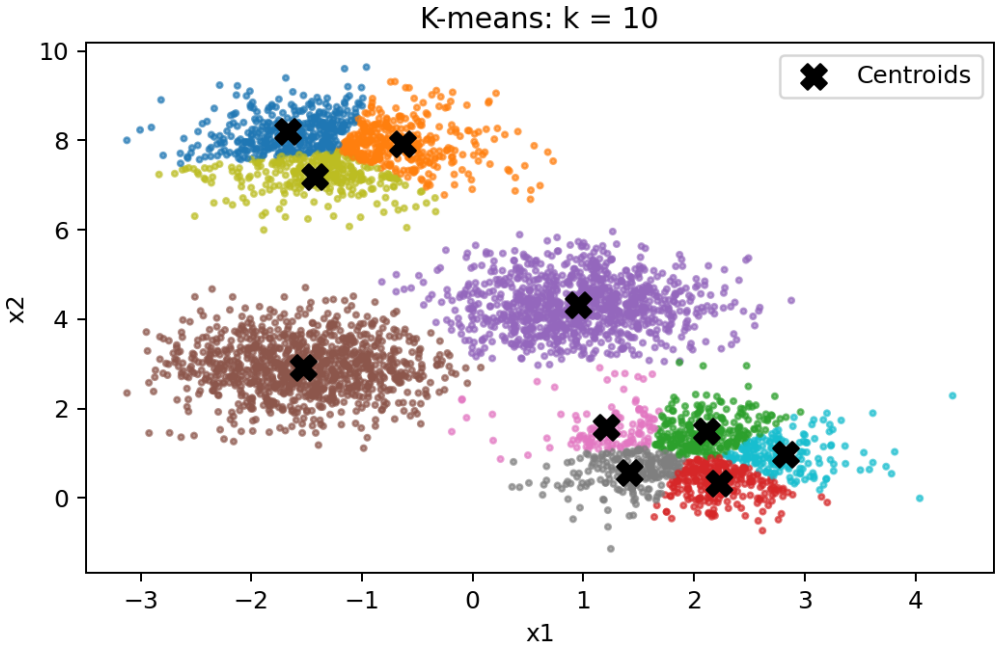
As k increases, coarse merged groups give way to finer partitions. With k=2 and 3 some generating groups must share a cluster. At k=4 the four generating groups are well recovered. At k=6 and 10, additional centroids subdivide groups. Lower squared error at larger k reflects added flexibility and does not imply more true generating groups. K-means can reach local optima and depends on initialization.




K-means: k = 4
Centroids
x1
x2


K-means: k = 6
Centroids
x1
x2


K-means: k = 10
Centroids
x1
x2
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
4

# 3. Noisy-label SVM and K-NN repair

Generate training and test sets independently, once, with 1,000 samples per class per set: positive N((1,1),I), negative N((-1,-1),3I). Keep features fixed for 20 corruption trials. Flip original positive labels with probability 0.35 and negatives with probability 0.20 using independent uniforms. Each paired comparison uses identical noisy labels. Both final classifiers are fresh RBF SVMs with C=1 and gamma=0.01.

Repair uses the five nearest other training rows, excluding self by infinite diagonal distance. Stable sorting resolves equal distances by training row index. Every vote reads the original noisy-label array; no clean labels or test information enter repair. Squared distances preserve Euclidean neighbor order.

Mean baseline accuracy: 0.827950 (82.795%). Mean repaired accuracy: 0.864375 (86.437%). Difference (repair minus baseline): +0.036425, or +3.643 percentage points.

For this seed, repair helped. Neighborhood voting can suppress isolated label errors when nearby examples share a class. It can also propagate errors near overlapping class regions or erase minority labels under asymmetric corruption. These results describe fixed train/test draws with repeated label noise; they do not establish performance over new dataset draws.

| Trial | Baseline | K-NN |
|---|---|---|
| 1 | 0.8345 | 0.8615 |
| 2 | 0.833 | 0.872 |
| 3 | 0.844 | 0.8675 |
| 4 | 0.8445 | 0.87 |
| 5 | 0.8395 | 0.8725 |
| 6 | 0.841 | 0.8715 |
| 7 | 0.8405 | 0.869 |
| 8 | 0.793 | 0.848 |
| 9 | 0.824 | 0.8615 |
| 10 | 0.8425 | 0.8715 |
| 11 | 0.822 | 0.86 |
| 12 | 0.819 | 0.8645 |
| 13 | 0.847 | 0.8645 |
| 14 | 0.8415 | 0.8715 |
| 15 | 0.816 | 0.8715 |
| 16 | 0.8285 | 0.8485 |
| 17 | 0.8235 | 0.8695 |
| 18 | 0.843 | 0.87 |
| 19 | 0.784 | 0.8335 |
| 20 | 0.798 | 0.869 |

## Source

Assignment: https://github.com/ucsc-cse-242/assignments/tree/main/Assignment3